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Research papers

Novel multi-layer nano-modified PCM configuration for efficient thermal management of photovoltaic-thermal systems

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ABSTRACT

Effective thermal management of photovoltaic (PV) systems is crucial for mitigating temperature-induced efficiency losses and enhancing overall performance. This study proposes a novel approach to address this challenge by employing multi-layer phase change materials (PCMs) with strategically degraded nanoparticle concentrations integrated into photovoltaic-thermal (PVT) systems. The transient heat-transfer dynamics, temperature distributions, and conversion efficiencies were numerically investigated for three configurations: a uniform single-layer nano-PCM, a two-layer nano-PCM with degraded concentrations, and a three-layer nano-PCM with degraded nanoparticle concentrations. Notably, the three-layer degraded configuration consistently outperformed other arrangements, maintaining PV temperatures up to 5 °C lower and electrical efficiencies up to 4 % higher over 100 min of operation. This configuration also achieved the highest thermal efficiency of 0.682, surpassing the single PCM case by 6.7 %. By combining multi-layer PCMs with degraded nanoparticle loadings, the proposed approach enabled more uniform heat distribution within the nano-PCM composite, overcoming the inherent limitation of decreasing thermal interaction between nanoparticles and PCM regions farther from the heat source observed in uniform nano-PCMs. The proposed approach for thermal management can significantly enhance the heat absorption, temperature regulation, and thermoelectric performance of PVT systems under continuous operation scenarios.

1. Introduction

The world's rising energy demands, coupled with the harmful environmental impacts of fossil fuels, have accentuated the critical need for transitioning to more sustainable and alternative energy sources [1]. Solar photovoltaic (PV) technology has emerged as a promising solution to meet these challenges by directly converting sunlight into electricity with zero emissions. However, the inherently low efficiency of commercial PV modules, rarely exceeding 20 % [2,3], remains a major limitation. A substantial portion of incident solar energy is converted to waste heat rather than electricity, adversely impacting the energy conversion process and causing performance degradation, especially in hot climates [4,5]. Effective thermal management strategies are therefore essential to mitigate temperature-induced efficiency losses in PV systems. Photovoltaic-thermal (PVT) technology, which integrates solar PV modules with a heat recovery system, has garnered significant interest as it simultaneously generates electricity and captures waste heat, enhancing overall system efficiency and cost-effectiveness [6].

The integration of phase change materials (PCMs) into PVT systems is considered an effective thermal management strategy to enhance performance and mitigate temperature-induced degradation of PV cells [7–9]. By incorporating PCMs within the PVT system, the excess heat generated by the PV cells can be absorbed during the melting phase of PCM, thereby reducing the operating temperature of the cells [10]. Conversely, during the solidification process, the stored latent heat can

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Nomenclature

A_{mush}	Mushy zone constant
A_{pv}	Front surface area of PV module (m^2)
C_p	Specific heat capacity (J/kg.K)
d_{np}	Nanoparticle diameter (m)
F	View factor
G_{solar}	Incident solar radiation flux (W/m^2)
g	Gravitational acceleration (m/s^2)
h	Heat transfer coefficient (W/m^2K)
H	Total enthalpy (J/kg)
h_{ref}	Reference enthalpy (J/kg)
k	Thermal conductivity (W/mK)
L_f	Latent heat of fusion (J/kg)
m_{pcm}	Mass of PCM (kg)
Nu	Nusselt number
p	Pressure (Pa)
Pr	Prandtl number
q_{sky}	Radiative heat loss (W/m^2)
$\dot{Q}_{charge}$	Rate of energy charge (W)
Ra	Rayleigh number
Re	Reynolds number
T	Temperature (K)
T_{amb}	Ambient temperature (K)
T_{init}	Initial temperature (K)
T_l	Liquidus temperature (K)
T_{ref}	Reference temperature (K)
T_s	Solidus temperature (K)
T_{sky}	Effective sky temperature (K)
t	Time (s)
V	Velocity vector (m/s)

Greek symbols

β	Thermal expansion coefficient (1/K)
β_k	Boltzmann correlation parameter
δ	Computational constant|
ε	Emissivity
η	Electrical efficiency
η_{ref}	Reference electrical efficiency
η_{Th}	Thermal efficiency
λ	Liquid fraction
μ	Dynamic viscosity (Pa.s)
ρ	Density (kg/m^3)
σ	Stefan-Boltzmann constant
ζ	Nanoparticle distribution factor
φ	Nanoparticle volume fraction

Abbreviations

NP	Nanoparticle
PCM	Phase change material
PV	Photovoltaic
PVT	Photovoltaic-thermal

be recovered and utilized for applications such as space heating or domestic hot water production [11,12]. Numerous experimental and computational studies have investigated the performance enhancement of PVT systems through PCM integration. Islam et al. [13] reported that a hybrid PVT-PCM unit offers 33–50 % higher thermal energy storage capability and 75–100 % prolonged heat energy disposal time compared to a conventional PVT collector. Browne et al. [14] and Preet et al. [15] observed improved electricity generation and temperature regulation with PCM integration. Computational studies by Imam et al. [16] and Su et al. [17] demonstrated enhanced thermal and overall efficiencies through numerical modeling of PVT-PCM systems.

While PCM integration has shown significant potential, a major limitation of conventional PCMs is their inherently low thermal conductivity, which can impede heat transfer rates during heat charging and discharging processes [18–20]. To overcome this limitation, researchers have explored incorporating highly conductive nanoparticles within the PCM to form nano-enhanced PCM (nano-PCM) composites [21,22]. Several studies have investigated the performance of PVT systems integrated with nano-PCMs. Selimefendigil and Şirin [23] experimentally studied a hybrid PVT-air collector with a CuO-based nano-PCM, achieving an overall efficiency of up to 82.19 %. Chibani et al. [24] numerically analyzed a PVT module integrated with nano-PCMs containing various nanoparticles, including MgO, TiO_2, ZnO, and CuO, indicating the potential for maintaining higher electrical efficiencies compared to conventional PCMs. While these studies demonstrated the benefits of nano-PCMs in enhancing the thermal performance of PVT systems, a limitation arises from the decreasing thermal interaction between the nanoparticles and the PCM regions farther away from the thermally active wall in uniform-concentration nano-PCMs. To address this limitation, researchers have explored alternative strategies, such as employing multi-layer PCMs with different melting points [25] or incorporating heat transfer enhancement techniques like finned structures [26], metallic foams [27], and porous matrices [28].

The concept of employing multi-layer PCMs in thermal energy management applications has been extensively studied, as it offers enhanced energy storage capacity and improved temperature regulation compared to single-layer PCM configurations [29]. Fang and Chen [30] investigated the effects of different multi-layer PCMs on thermal functionality of a latent-heat storage unit, while Liu et al. [31] studied an air-multi-layer PCMs unit for free cooling and ventilation. Mahdi et al. [32,33] explored solidification enhancement with multi-layer PCMs, cascaded metal foams, and nanoparticles in shell-and-tube energy storage systems. However, the proposed approach in this study combines the use of multi-layer PCMs with degraded nanoparticle concentrations, a novel technique that has not been extensively investigated in the context of PVT systems.

This study aims to address the limitations of uniform nano-PCMs by proposing a novel multi-layer configuration with strategically degraded nanoparticle concentrations. It is a novel approach in which the concentration of nanoparticles is gradually reduced across the layers of multiple PCMs. Specifically, higher nanoparticle concentrations are placed in the layers closer to the heat source (the PV module), and the concentration degrades along the heat flow direction. By incorporating multi-layer PCMs, each layer contains a unique nanoparticle concentration along with PCM having a relevant melting point, better thermal management by PVT is allowed. The multi-layer nano-PCM configuration is expected to improve heat communication throughout the PCM layers, addressing the decreased thermal interaction in regions far from the heat source (the PV module) that occurs with uniform nano-PCMs. Consequently, the overall thermal management capabilities of the PVT system could be significantly enhanced, potentially resulting in improved energy storage and recovery rates, as well as better temperature regulation of the PV cells. Through a comprehensive numerical investigation, this study aims to evaluate the potential benefits of the proposed approach by analyzing the heat transfer dynamics, temperature distribution, and overall thermal management potential of the PVT system during the energy charge mode. The outcomes of this study could provide valuable insights into optimizing the thermal management capabilities of PVT systems and potentially pave the way for broader applications in various thermal energy storage and management scenarios. However, it is important to note that the effectiveness of this approach may be subject to certain limitations depending on specific system

parameters and operating conditions, which will be discussed in detail in the results section.

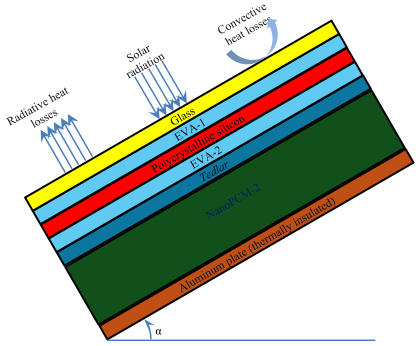
2. System and problem statement

The PVT system under consideration consists of a PV module vertically mounted onto a rectangular multi-segment casing that houses the nano-PCM composites (Fig. 1). The PV module and the PCM casing are thermally interconnected to facilitate efficient conductive heat transfer from the PV cells to the PCMs during operation. This design allows the heat generated by the PV cells to be absorbed through the melting process of the PCMs, consequently regulating the operating temperature of the cells and mitigating potential performance degradation due to excessive temperature buildup. The PV panel is a conventional flat-plate configuration comprising multi-layer layers as follows (from top to bottom): 3 mm tempered glass cover, 0.5 mm ethylene-vinyl acetate (EVA-1) encapsulant, 0.2 mm polycrystalline silicon cells, 0.5 mm EVA-2 encapsulant, and a 0.3 mm Tedlar sheet [34]. The thermophysical properties of these materials, as well as those of the selected PCMs (RT26, RT31, and RT35 supplied by Rubitherm [35]) and the 2-mm aluminum plates, are summarized in Table 1. The PCMs have melting temperature ranges of 298–299 K for RT26, 301–307 K for RT31, and 302–309 K for RT35. To form the nano-PCM composites, three nanoparticle types were considered: alumina, copper, and graphene (properties summarized in Table 2).

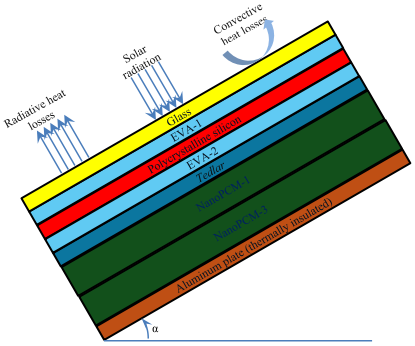
The arrangement of PCM layers and nanoparticle concentrations was carefully designed to optimize thermal management of the PVT system. Three PCMs (PCM-1: RT26, PCM-2: RT31, and PCM-3: RT35) with strategically distributed melting points were selected to manage a wide range of temperature fluctuations. The PCM layers were arranged from lowest to highest melting point (PCM-1 to PCM-3) to mimic the natural temperature variation profile within the system. As temperature rises gradually across the PCM domain with decreasing distance from the heat source (i.e., PV cells), this layering approach allows effective heat absorption at different temperature stages, improving thermal distribution uniformity. Effective initial heat absorption is critical to prevent early overheating of the PV cells. PCM-1, with the lowest melting point, is positioned closest to the cells to rapidly buffer any generated heat. Its proximal location ensures a prompt thermal response, which is crucial for maintaining optimal PV cell temperature. In contrast, the higher melting point of PCM-3 makes it less suitable for this immediate heat-sinking role close to the cells, hence its placement furthest from the heat source.

The nanoparticle concentrations were designed with a degraded profile, featuring higher concentrations in the top layer to enhance initial heat absorption and distribution near the heat source, with a gradual decrease through the middle and bottom layers for more controlled heat release. Three distinct configurations were developed, as illustrated in Fig. 1: (a) a single nano-PCM layer with a uniform nanoparticle concentration of 0.05 throughout, serving as the reference case; (b) a two-layer arrangement with the top layer having a nanoparticle concentration of 0.10 closest to Tedlar sheet, and the bottom layer having a concentration of 0.00; (c) a three-layer arrangement with the top layer having a nanoparticle concentration of 0.10 closest to Tedlar sheet, the middle layer having 0.05, and the bottom layer having 0.00. The details of these nano-PCM arrangements, including the melting points and nanoparticle volume fractions in each layer, are provided in Table 3. In this context, higher nanoparticle content in PCM-1 enhances initial heat spreading near the PV cells, while the gradual reduction through lower layers optimizes heat transfer during melting. To validate this strategy, additional simulations were conducted comparing reversed PCM order from highest to lowest melting point. Results demonstrated the benefits of the proposed layered arrangement with degraded concentrations over the alternative arrangement.

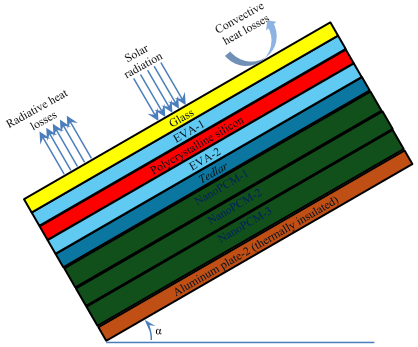
A consistent thickness of 21 mm was maintained for the whole PCM layer in all PVT configurations, resulting in 21 mm, 10.5 mm and 7 mm



(a) 1-PCM arrangement with 1-NP concentration



(b) 2-PCM arrangement with 2-NP concentrations



(c) 3-PCM arrangement with 3-NP concentrations

Fig. 1. Schematic diagrams of PVT module with multi-layer PCM arrangements and varying nanoparticle concentrations.

Table 1
The nominal thermophysical properties of the nanoPCM-based PVT system [35,37,38].

Property	PCM-1 (RT-25)	PCM-2 (RT-31)	PCM-3 (RT-35)	Glass	PV cell	EVA	Tedlar	Aluminum plates
C_p (J/kg K)	2000	2000	2000	500	677	2070	1250	870
ρ (kg/m^3)	760	760	770	3000	2330	960	1200	2700
k (W/m K)	0.2	0.2	0.2	2	148	0.3	0.2	202
Thickness (mm)	–	–	–	3	0.2	0.5	0.3	2
L [kJ/kg]	180	165	160	–	–	–	–	
T_s [K]	298	301	302	–	–	–	–	
T_l [K]	299	307	309	–	–	–	–	
β [1/K]	0.0005	0.0005	0.0005	–	–	–	–	

Table 2
Thermophysical properties of the nanoparticle manufacturing materials.

Property	Alumina	Copper	Graphene
ρ [kg/m^3]	3600	8920	2200
C_p [J/kg.K]	765	380	790
k [W/m.K]	36	400	5000

Table 3
Details of multi-layer PCM configurations and corresponding nanoparticle volumetric concentrations in the layered nano-PCM arrangements.

Figure #	No. of layers	Arrangement of the multi-layer PCMs	Melting points of the multi-layer PCMs (K)	NP volume-fraction arrangement
Fig. 1(a)	1	NanoPCM-2	306	0.05
Fig. 1(b)	2	NanoPCM-1|NanoPCM-3	299|309	0.10|0.00
Fig. 1(c)	3	NanoPCM-1|NanoPCM-2|NanoPCM-3	299|306|309	0.10|0.05|0.00

for single-layer, two-layer, and three-layer configurations, respectively. This approach was selected to isolate the effects of the degraded nanoparticle concentration on thermal management performance. By keeping layer thicknesses uniform, a near-constant mass across all configurations was achieved through careful study design. This was achieved via two key approaches: (1) PCMs with almost similar densities were selected (PCM-1: 760 kg/m^3, PCM-2: 760 kg/m^3, PCM-3: 770 kg/m^3, as shown in Table 1), and (2) a constant total PCM volume was maintained by using a consistent overall PCM thickness of 21 mm for all configurations. Through this approach, a maximum mass variation of less than 2.7 % between configurations was achieved, allowing for useful performance comparisons. The proposed hybrid approach, which combines multi-layer PCMs with degraded nanoparticle loadings, aims to enable more uniform heat distribution within the nano-PCM composite. This scheme seeks to overcome the inherent limitation of decreasing thermal interaction between nanoparticles and PCM regions farther from the heat source, a known issue with uniform-concentration nano-PCMs [36].

3. Mathematical formulation

3.1. Nano-PCM thermal model

The thermal behavior of the nano-PCM composite is modeled using a set of governing equations that describe the conservation of mass, momentum, and energy. These equations are solved numerically using the finite volume method and the enthalpy-porosity technique [39], which treats the mushy zone as a pseudo-porous region. So, the mass conservation equation is formulated as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{V}\right) = 0 \tag{1}$$

where ρ is the density, t is time, and $\vec{V}$ is the velocity vector.

The momentum conservation equation, considering the Boussinesq approximation and the Darcy-Brinkman term for the mushy zone, is given by:

$$\rho \frac{\partial \vec{V}}{\partial t} + \rho\left(\vec{V}.\vec{\nabla}\right)\vec{V} = -\nabla p + \mu \nabla^2 \vec{V} + \rho \vec{g}\left(T - T_{ref}\right) + \frac{(1-\lambda)^2}{\left(\lambda^3 + \delta\right)} A_{mushy} \vec{V} \tag{2}$$

where p is the pressure, μ is the dynamic viscosity, $\vec{g}$ is the gravitational acceleration, β is the thermal expansion coefficient, T is the temperature, T_{ref} is the reference temperature, λ is the liquid fraction, δ is a small computational constant (typically 0.001) to avoid division by zero, and A_{mush} is the mushy zone constant (typically 10^5–10^7) [40,41]. In this analysis, a value of 10^5 was found to provide good matching with experimental data from Huang et al. [42] used for validation.

The energy conservation equation, incorporating the latent heat of fusion, is expressed as:

$$\frac{\partial}{\partial t}(\rho H) + \nabla \cdot (\rho V H) = \nabla \cdot (k \nabla T) \tag{3}$$

Here H is the total enthalpy, and k is the thermal conductivity. The total enthalpy is calculated as the sum of sensible and latent heat contributions:

$$H = h_{ref} + \int_{T_{ref}}^{T} C_p dT + \lambda L_f \tag{4}$$

where h_{ref} refers to the enthalpy at T_{ref}, and L_f denotes the latent heat of fusion. The liquid fraction, λ, is updated at each iteration based on the temperature field using the following relationship:

$$\begin{aligned} \lambda &= \frac{T - T_s}{T_l - T_s} \quad if\ T_s \le T \le T_l \\ \lambda &= 0 \qquad if\ T < T_s \\ \lambda &= 1 \qquad if\ T > T_l \end{aligned} \tag{5}$$

where T_s and T_l are the solidus and liquidus temperatures, respectively.

The effective thermophysical properties of the nano-PCM composite are estimated using appropriate mixture models that account for the nanoparticle volume fraction and the properties of the base PCM and nanoparticles.

$$\rho_{npcm} = \varphi \rho_{np} + (1-\varphi)\rho_{pcm} \tag{6}$$

$$\left(\rho C_p\right)_{npcm} = \varphi\left(\rho C_p\right)_{np} + (1-\varphi)\left(\rho C_p\right)_{pcm} \tag{7}$$

$$(\rho L)_{npcm} = (1-\varphi)(\rho L)_{pcm} \tag{8}$$

$$(\rho \beta)_{npcm} = \varphi(\rho\beta)_{np} + (1-\varphi)(\rho\beta)_{pcm} \tag{9}$$

The effective dynamic viscosity and thermal conductivity of the composite are evaluated as [43]:

$$\mu_{npcm} = 0.983\, e^{(12.959\varphi)} \mu_{pcm} \tag{10}$$

$$k_{npcm} = \frac{k_{np} + 2k_{pcm} - 2(k_{pcm} - k_{np})\varphi}{k_{np} + 2k_{pcm} + (k_{pcm} - k_{np})\varphi} k_{pcm} + 5 \times 10^4 \beta_k \zeta \varphi \rho_{pcm} C_{p,pcm} \sqrt{\frac{BT}{\rho_{np} d_{np}}} (2.8217 \times 10^{-2}\varphi + 3.917 \times 10^{-3}) \frac{1}{T_{ref}} + (-3.0669 \times 10^{-2}\varphi - 3.91123 \times 10^{-3}) \quad (11)$$

where: β_k is Boltzmann constant, 1.381×10^{-23} J/K, and $B = 8.4407(100\varphi)^{-1.07304}$.

This nano-PCM thermal model, coupled with the appropriate initial and boundary conditions, enables the simulation of the melting process and the subsequent heat transfer within the PVT system, accounting for the presence of nanoparticles and their influence on the thermal behavior of the PCM composite.

3.2. PV thermal model

The thermal behavior of PV cell in the PVT system is modeled by considering the energy balance at the top surface of the module. The energy balance equation accounts for the incident solar radiation, thermal losses due to convection and radiation, and the heat transfer to the underlying PCM layer. The convective heat transfer coefficient is determined based on the flow regime, accounting for both natural and forced convection.

The energy balance at the top surface of the PV module can be expressed as [25]:

$$-k_{glass}\frac{\partial T}{\partial y} = G_{solar}(\alpha_e - \eta) - (h_{forced} + h_{free})(T_{glass} - T_{amb}) - q_{sky} \quad (12)$$

The PV conversion efficiency is then calculated based on the obtained temperature distribution, using an empirical relationship that accounts for the temperature dependence:

$$\eta = \eta_{ref} \cdot \{1 - \beta_{ref}(T_{PV} - T_{ref})\} \quad (13)$$

The radiative heat dissipation from the collector's top surface to the sky can be formulated as:

$$q_{sky} = \epsilon_{glass} F \sigma \left(T_{glass}^4 - T_{sky}^4\right) \quad (14)$$

where ε_{glass} is the emissivity of the glass cover, σ is the Stefan-Boltzmann constant, and T_{sky} is the effective sky temperature, which is estimated using empirical correlations based on the ambient temperature (T_{amb}).

$$T_{sky} = 0.037536 T_{amb}^{1.5} + 0.32 T_{amb} \quad (15)$$

The convective heat loss is calculated using the Newton's law of cooling where the natural and forced convective heat dissipations from the collector's top surface are evaluated by estimating the heat transfer coefficients, h_{free} and h_{forced}, as:

$$h_{natural} = \begin{cases} \left(0.67(Cos\alpha)Ra^{0.25} / \left((0.492k/\mu C_p)^{9/16} + 1\right)^{0.44} + 0.68\right)\frac{k}{\ell}, & Ra < 10^9 \\ \left(0.387 Ra_l^{0.167} / \left((0.492k/\mu C_p)^{9/16} + 1\right)^{0.44} + 0.825\right)\frac{k}{\ell}, & Ra \geq 10^9 \end{cases} \quad (16)$$

where Rayleigh number $\left(Ra = \frac{g\beta_{air}\ell^3\rho^2 C_p}{k\mu}(T_p - T_{amb})\right)$

$$h_{forced} = \begin{cases} \left(0.3387 Pr^{1/3} Re^{1/2} / \left(1 + ((0.0468/Pr)Re^{2/3})^{1/4}\right)\right)\frac{2k}{\ell}, & Re < 10^5 \\ \left(Pr^{1/3}(0.037 Re^{4/5} - 871)\right)\frac{2k}{\ell}, & Re \geq 10^5 \end{cases} \quad (17)$$

where Reynolds number $\left(Re = \frac{\rho V \ell}{\mu}\right)$ and Prandtl number $\left(Pr = \frac{C_p \mu}{k}\right)$.

3.3. Initial and boundary conditions

3.3.1. Initial conditions

- The PVT system is initially at a uniform temperature, T_0, representing the ambient temperature at the start of the simulation.
- The liquid fraction of the PCM is initially set to zero, indicating a fully solid state.
- The velocity field within the PCM domain is initially set to zero, representing a stationary condition.

3.3.2. Boundary conditions

- A constant solar radiation flux, G_{solar}, is applied to the top surface of the PV module, representing the incident solar energy. The convective and radiative heat losses from this surface are calculated based on the ambient conditions, as described in the PV thermal model section.
- The interface between the PV backsheet and the adjacent PCM layer is assumed thermally-coupled boundary condition. The heat flux from the PV module to the PCM is estimated based on the temperature difference and the effective thermal conductivity of the intervening layers.

$$k_{PCM}\, \partial T_{(i)} / \partial n = k_{pv}\, \partial T_{(i)} / \partial n \quad (18)$$

where i is an index showing the interface between he PV backsheet and the adjacent PCM layer, k_{pv} is the effective thermal conductivity of the intervening PV layers (EVA and Tedlar), and $\partial T/\partial n$ is the temperature gradient normal to the interface for each material.

- The lateral boundaries of the PCM domain are considered adiabatic, assuming negligible heat transfer through the side walls. The bottom boundary of the PCM domain is also considered adiabatic, assuming perfect insulation at the base.
- At the interface between the liquid PCM and the solid boundaries (e.g., aluminum plates), the no-slip boundary condition is applied for the velocity field, and the temperature is set to be continuous across the interface.

These initial and boundary conditions are implemented in the numerical model through appropriate techniques, such as user-defined functions or boundary condition specifications in the computational fluid dynamics (CFD) solver.

3.4. The energy charge rate and thermal efficiency

The rate of energy charge is a crucial parameter that quantifies the thermal functionality of the PVT system during the melting process of the PCM. In the case of a multi-layered PCM configuration, where different PCMs with varying melting temperatures are employed, the rate of energy charge can be calculated as the sum of the individual contributions from each PCM layer:

$$\dot{Q}_{charge} = \sum_{i=1}^{n} (m_{pcm})_i \left((C_{p,s})_{pcm} (T_{init} - T_l) + L_f + (C_{p,l})_{pcm} (T_s - T_{end}) \right)_i / t_m \quad (19)$$

where the subscript 'i' denotes the individual PCM layer, and the summation is performed over all PCM layers present in the system.

Thermal efficiency is another crucial parameter for quantifying the thermal functionality of PVT systems and comparing the effectiveness of different design configurations or operating conditions.

$$\eta_{Th} = \dot{Q}_{charge} / (G_{solar} A_{pv}) \quad (20)$$

Here, A_{pv} is the front surface area of the PV module.

A higher thermal efficiency indicates a more rapid absorption of thermal energy by the PCM, which can lead to better temperature regulation and potential improvements in the overall efficiency of the PVT system.

4. Numerical procedure and validation

The numerical simulations were carried out using the commercial computational-fluid-dynamics solver (ANSYS Fluent 24R1). The finite volume method was employed to discretize the governing equations, and the enthalpy-porosity technique was utilized to model the phase change process. The pressure-velocity coupling was handled using SIMPLE algorithm [44]. The transient simulations were conducted using an appropriate time step size determined through a time step independence study to accurately capture the phase change dynamics. The transient heat-transfer dynamics, temperature distributions, and conversion efficiencies were numerically investigated for three configurations: a uniform single-layer nano-PCM, a two-layer nano-PCM with degraded nanoparticle concentrations, and a three-layer nano-PCM with degraded nanoparticle concentrations. To accurately model the phase change process, the enthalpy-porosity technique was utilized. This method treats the liquid-solid mushy zone as a porous medium, allowing for a smooth transition between solid and liquid states. The simulations provided insights into the temperature evolution, phase change dynamics, and overall thermal management performance of the different configurations over time.

A comprehensive grid independence study was conducted by evaluating three different mesh sizes: 0.10, 0.50, and 1.00 mm. The simulations were performed for each mesh, and the results were compared in terms of accuracy and computational time. Ultimately, a mesh size of 0.50 mm was selected as it provided a good balance between solution accuracy and reasonable simulation times. Further increasing the mesh resolution did not yield significant improvements in the solution, as depicted in the grid independence plot (Fig. 2(a)). In addition to the spatial discretization, a time step independence study was also performed. Time step sizes of 0.05 s, 0.1 s, and 0.15 s were evaluated (Fig. 2 (b)), and the results were analyzed in terms of the convergence and stability of the solution. Based on the findings, a time step size of 0.1 s was chosen as it demonstrated satisfactory convergence and stability while allowing for reasonable computational times during the simulations of the melting process. The convergence criteria were set to 10^{-4}

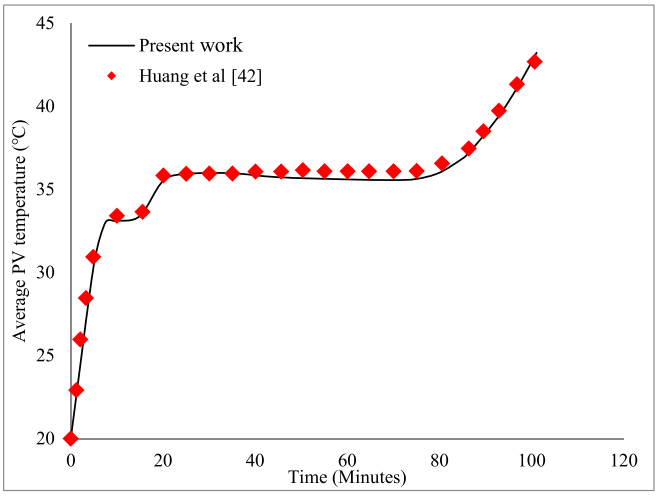


Fig. 3. Comparative validation of the present numerical results against Huang et al.'s experimental data [42].

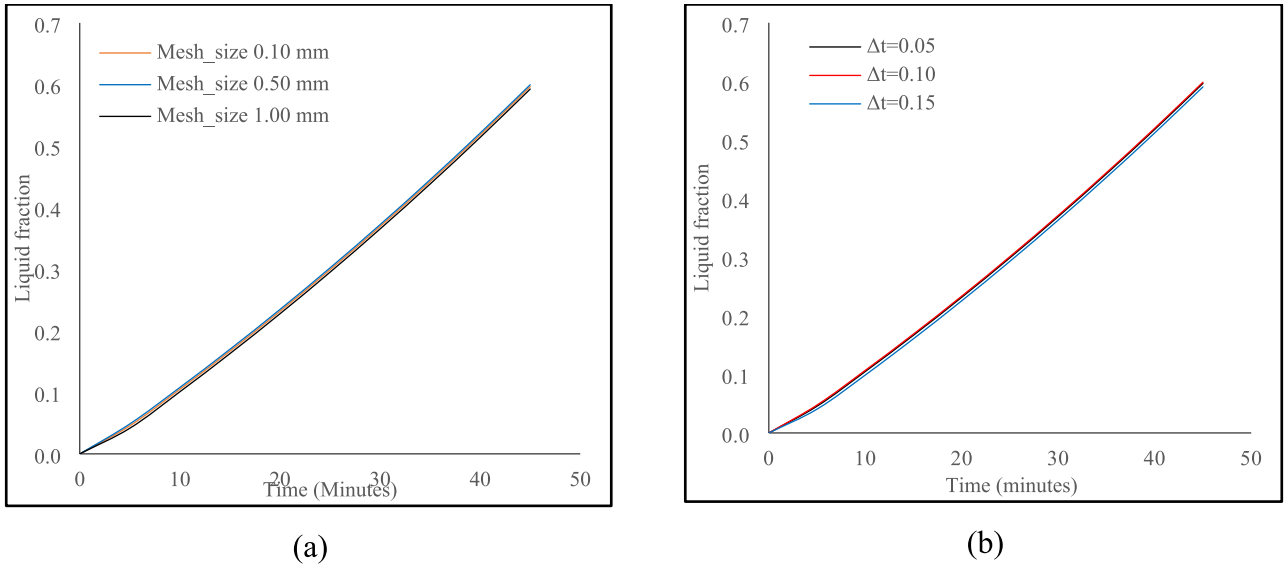


Fig. 2. Effects of (a) mesh grid size and (b) timestep size on transient liquid fraction during PCM melting.

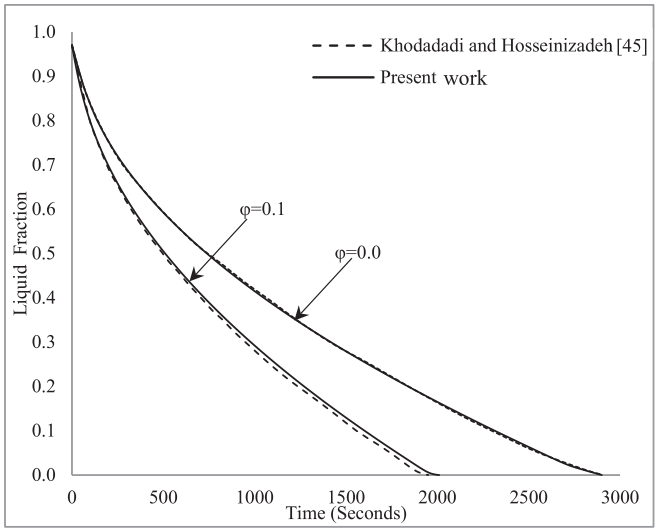


Fig. 4. Validation of transient liquid-fraction results with Khodadadi and Hosseinizadeh [45].

for the continuity and momentum equations, and 10^{-6} for the energy equation. The computational domain was discretized using a structured mesh with a grid size of 0.5 mm, determined through a mesh independence study to ensure grid-independent results while maintaining computational efficiency. The under-relaxation factors for the velocity components, pressure, and thermal energy were set to 0.5, 0.3, and 1.0, respectively, to promote convergence stability.

The model was validated via established data from two independent sources to ensure its accuracy and reliability. First, the transient temperature profiles obtained from the present simulations were compared with the experimental results reported by Huang et al. [42] for a PVT unit featuring a PCM layer, as shown in Fig. 3. The match between the predicted and measured temperatures confirmed the model's capability to accurately capture the thermal behavior of the PVT system during the melting process. Additionally, the model was validated against the numerical findings of Khodadadi and Hosseinizadeh [45] in Fig. 4 for the melting of a nano-PCM composite within a differentially heated enclosure. The transient liquid fraction profiles obtained from the present simulations showed close agreement with the cited results, further validating the model's accuracy in simulating the phase change process

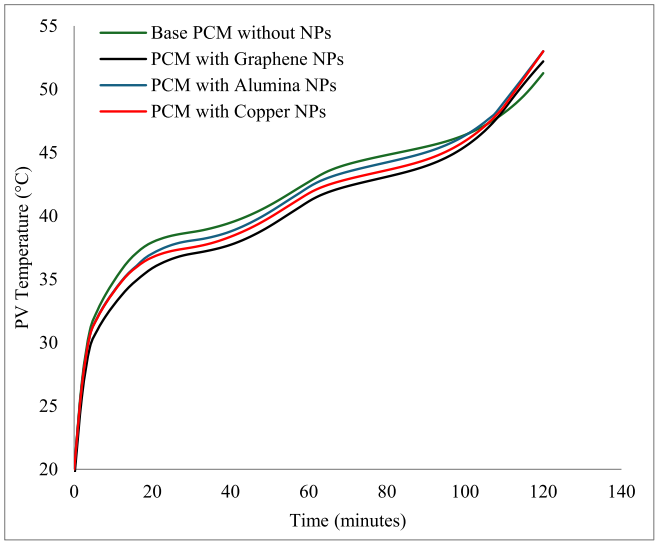


Fig. 5. Transient average temperature profiles of PV module with graphene, alumina, copper nano-PCM composites and base PCM without NPs.

and the influence of nanoparticles on the thermal behavior of the PCM composite. With the numerical model validated against both experimental and numerical data from independent sources, confidence is obtained in its ability to reliably predict the thermal behavior of the proposed multi-layer nano-PCM configurations in the PVT system.

5. Results and discussion

5.1. Impact of NP material on performance of nanoPCM-based PVT

This study examined the performance of different nano-PCM composites using three types of nanoparticles (NPs): graphene, alumina, and copper, as well as the base PCM without any NPs. While graphene was selected as the primary focus due to its superior thermophysical properties, a comprehensive comparison was conducted to evaluate the relative performance of each material.

Fig. 5 provides a comparison of the average temperature of the PV module integrated with different nano-PCM composites over a period of 120 min. Initially at time zero, all cases exhibit similar average temperature of around 20 °C. As heat generation starts in the PV module, its temperature begins to rise gradually over time for all configurations. At 60 min, it can be seen that the graphene nano-PCM has successfully achieved the lowest average temperature of 40.1 °C among all the cases. In comparison, the base PCM without nanoparticles results in a notably higher temperature of 42.9 °C. This demonstrates the superior thermal regulation capability of the graphene nano-PCM. The alumina and copper nano-PCMs also outperform the base PCM, but to a lesser extent than the graphene nano-PCM. Specifically, at 60 min their average temperatures are 42.2 °C and 41.9 °C respectively, still higher than the graphene nano-PCM but lower than the base PCM temperature. As time progresses to 100 min, the superior thermal performance of the graphene nano-PCM remains consistent, maintaining the lowest average temperature of 51.4 °C. In contrast, the alumina and copper nano-PCMs show higher temperatures of 53.2 °C and 53.7 °C respectively.

Fig. 6 provides a comparison of the electrical efficiency of the PV module integrated with different nano-PCM composites over 120-minute duration. Of the various nanoparticle materials evaluated, the graphene nano-PCM exhibits the highest electrical efficiency throughout the time range studied, demonstrating its exceptional thermophysical properties. At the beginning of the simulation, all the cases show similar electrical efficiency around 0.165 since little heat is generated. As time progresses, the temperature of the PV module starts to rise slowly due to the continuous heat generation. At 60 min, while the electrical efficiency of the base PCM drops to 0.156, the graphene nano-PCM is able to maintain the highest efficiency of 0.163, an improvement of 5.6 % compared to the base case. This can be attributed to the graphene nanoparticles' high thermal conductivity, which facilitates rapid heat transfer within the composite and prevents a sharp temperature rise of the PV module. The alumina and copper nano-PCMs also exhibit better electrical efficiencies than the base PCM, but not as high as the graphene nano-PCM. Specifically, at 60 min, their efficiencies are 0.159 and 0.158 respectively, representing only 1.9 % and 1.3 % improvement over the base PCM.

As time advances to 100 min, the graphene nano-PCM maintains a higher electrical efficiency (0.156) compared to the base PCM (0.150), demonstrating a 4.1 % improvement. However, by 120 min, this advantage diminishes, with all cases converging around 0.150 efficiency. Interestingly, the base PCM without NPs slightly outperforms the nano-PCM cases at this final time point. This behavior highlights the dynamic nature of PCM-based thermal management in PVT systems. The mid-term superior performance of nano-PCMs is attributed to enhanced PCM melting dynamics. However, as the PCM approaches melting completion, its thermal regulation capacity decreases, leading to slight reversal in performance as observed at the 120-minute mark.

These results demonstrate the superior performance of graphene nanoparticles in enhancing both thermal regulation and electrical

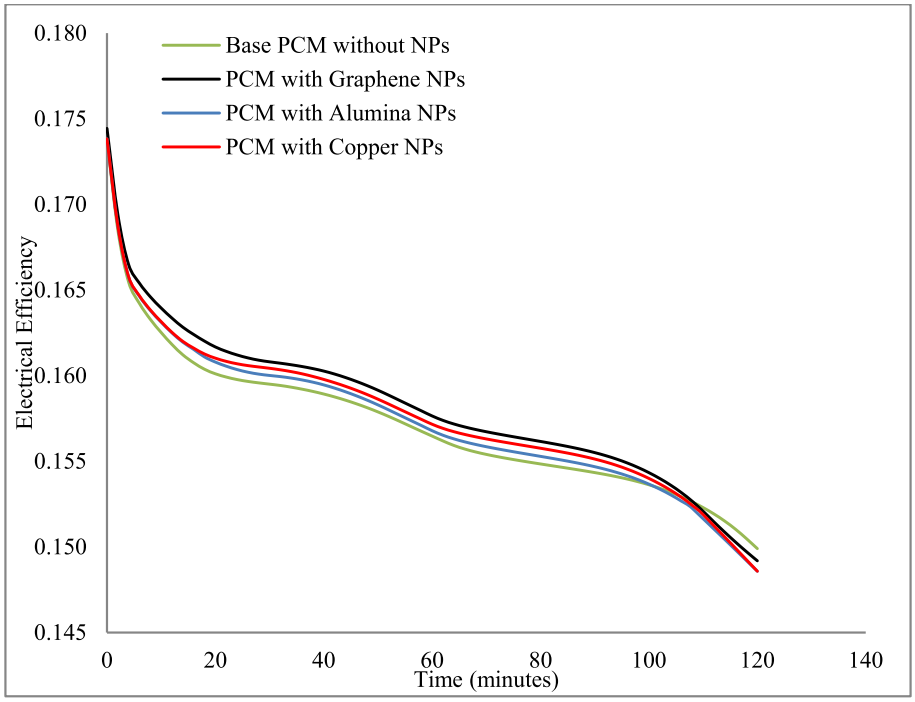


Fig. 6. Transient electrical efficiency profiles of PV module with graphene, alumina, copper nano-PCM composites and base PCM without NPs.

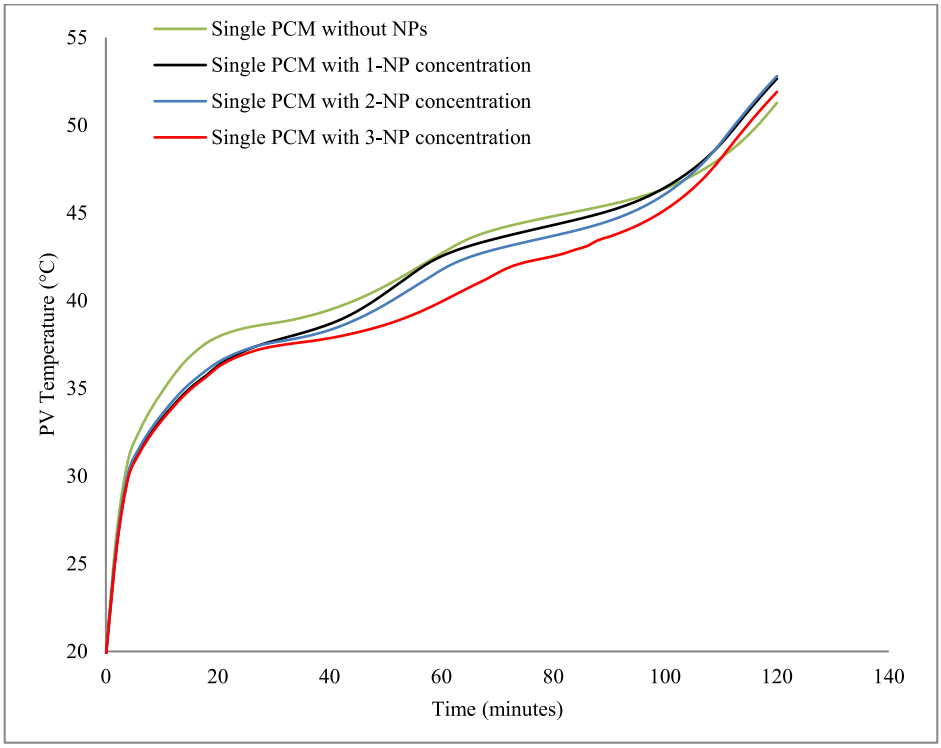


Fig. 7. Transient PV temperature profiles for single PCM with uniform and degraded nanoparticle concentrations.

efficiency of the PVT system, particularly in the initial and mid-term periods. The exceptionally high thermal conductivity of graphene allows for more rapid and uniform heat distribution within the PCM, effectively preventing localized temperature spikes and maintaining good operating conditions for the PVT during these crucial periods. While alumina and copper nanoparticles also show improvements over the base PCM, their impact is less pronounced compared to graphene.

5.2. Employing nano-PCM with degraded nanoparticle concentration in PVT

Fig. 7 compares the transient PV temperature profiles for the three different nano-PCM arrangements considered in the study at various time intervals during the melting process. At the initial time of 20 min, all the cases exhibit similar average PV temperature of around 30 °C,

since little heat has been generated at this point. As time progresses to 40 min, the single layer arrangement with uniform nanoparticle concentration starts experiencing a rise in temperature, reaching 34 °C. In comparison, the 2-layer arrangement with degraded concentrations remains slightly lower at 33 °C, indicating faster heat diffusion. Notably, the 3-layer arrangement shows the best thermal performance with the lowest temperature of 32 °C at this stage, demonstrating the benefits of degraded concentrations of nanoparticles and multi-layer PCM layers. As melting continues and more heat is absorbed, the differences in thermal regulation capabilities among the arrangements become more discernible. At 60 min, the uniform concentration arrangement lags behind with a PV temperature of 38 °C, while the 2-layer arrangement is still lower at 36.5 °C. The 3-layer arrangement continues displaying superior thermal control, maintaining the average temperature significantly lower at 35 °C. A similar trend persists at the later time of 80 min, with the uniform, 2-layer, and 3-layer configurations recording temperatures of 41 °C, 39.5 °C, and 38 °C respectively. This clearly highlights the gradual increase in thermal performance obtained by strategically employing degraded nanoparticle concentrations within PCM layers.

As time advances to around 100 min, the 3-layer degraded arrangement demonstrates superior thermal regulation, with average PV temperatures for 1-layer, 2-layer, and 3-layer cases at 46 °C, 44 °C, and 43 °C respectively. This shows the effectiveness of the 3-layer degraded arrangement in maintaining lower temperatures over an extended period. However, by 120 min, an important shift is observed: the Single PCM without NPs case exhibits the lowest average PV temperature. This result highlights the complex thermal behavior of PVT systems over extended periods. It suggests that while nanoparticle-enhanced and multi-layer PCMs offer advantages in initial and mid-term thermal management, their rapid melting completion may lead to reduced effectiveness in later stages compared to a single, unenhanced PCM layer. This behavior can be attributed to the different melting rates and thermal storage capacities of the various configurations. The nano-enhanced PCMs, due to their improved thermal conductivity, melt more quickly and thus may exhaust their heat absorption capacity earlier. In contrast, the single PCM without NPs melts more slowly, potentially providing a more sustained cooling effect over a longer period.

Fig. 8 compares the transient electrical efficiency profiles of the PV system for the different nano-PCM arrangements over the melting period of 120 min. At the beginning, all cases exhibit similar initial electrical efficiency of around 17.5 % since the temperature is uniform. As heat generation proceeds, the efficiency begins to decline slowly for all configurations due to rising temperatures affecting performance. By 60 min, the distinctions in thermal management capabilities of the arrangements become evident. The single uniform concentration arrangement results in the lowest efficiency of 16.2 % owing to poorer heat transfer properties. In contrast, the 2-layer degraded concentration arrangement displays marginally better efficiency of 16.4 % as temperatures remain somewhat regulated. Impressively, the 3-layer degraded arrangement continues exhibiting the best thermal behavior, upholding the highest efficiency of 16.6 % among all cases at this point.

As melting continues further, the deviations in electrical performance keep augmenting steadily. By around 100 min, the 3-layer arrangement with degraded nanoparticle concentrations demonstrates superior performance, maintaining an electrical efficiency of 16.1 % compared to 15.7 % for the 2-layer and 15.5 % for the uniform case. However, at 120 min, an interesting shift in performance is observed. The multi-PCM alone case exhibits the highest electrical efficiency at this time point, outperforming the nano-enhanced configurations. This shift in performance likely occurs due to the different melting rates and thermal storage capacities of the various configurations. While the 3-layer degraded NP arrangement offers superior performance for a significant portion of the melting process, the single PCM without NPs may be more effective at the very late periods of process. In summary, the degraded configuration successfully maintains a lower operating temperature for the PV module till the point of PCM melting completion. Thereby, it upholds electrical efficiency gains compared to the other arrangements throughout the entire PCM melting period.

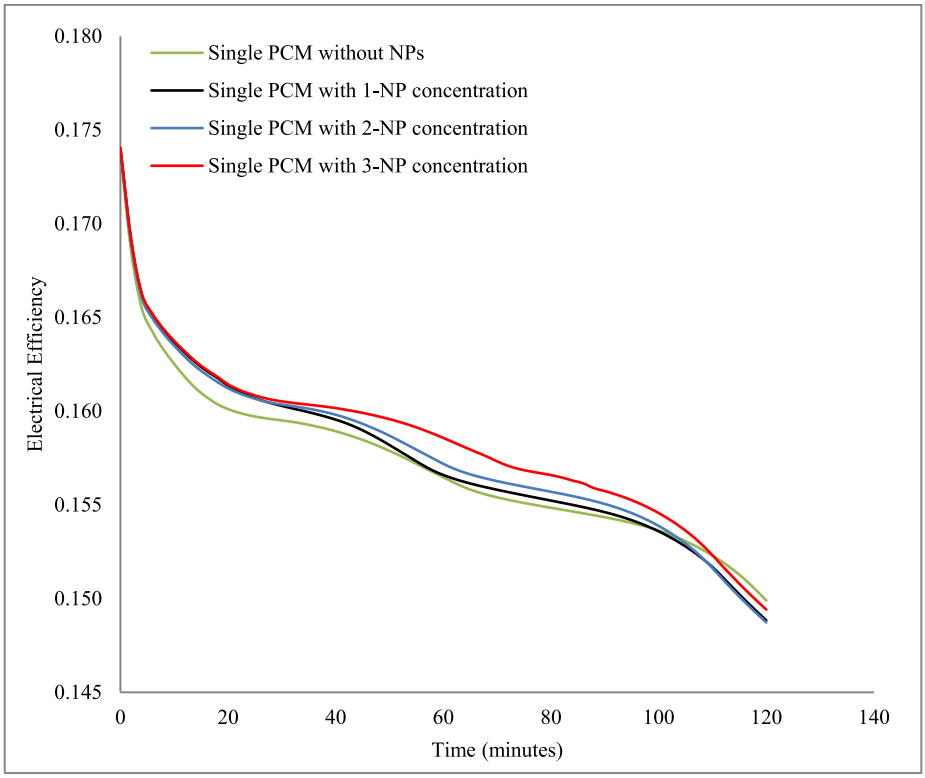


Fig. 8. Transient electrical efficiency profiles of PV module for single PCM with uniform and degraded nanoparticle concentrations.

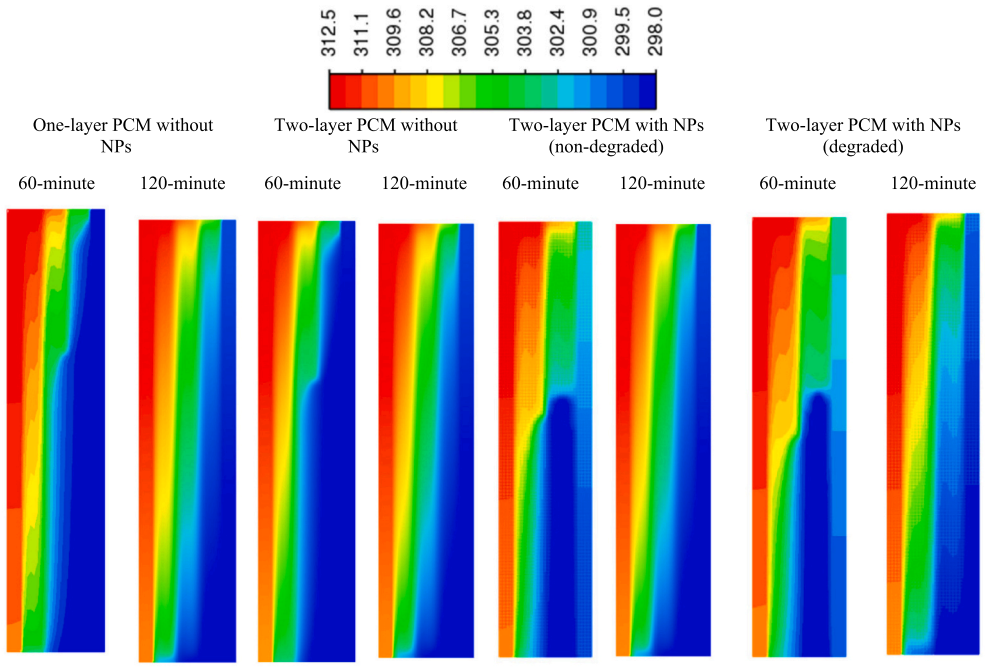


Fig. 9. Temperature distribution contours of PVT system for three-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

5.3. Employing multi-layer nano-PCMs with degraded nanoparticle concentration in PVT

Fig. 9 shows the temperature distribution contours of the PVT system for three different configurations: a single-layer PCM without NPs, a two-layer PCM without NPs, and two cases involving a two-layer PCM with NPs – one with a non-degraded arrangement and another with a degraded arrangement of NP concentrations. The contour plots are shown at two different time instants of 60 and 120 min. At the 60-minute mark, the single-layer PCM case exhibits a relatively higher temperature region near the heat source (PV module), indicating localized heat accumulation. The two-layer PCM configuration without NPs shows a more uniform temperature distribution compared to the single-layer case, suggesting improved heat dissipation due to the presence of multiple PCMs with different melting points. However, the two-layer PCM with non-degraded NP concentrations displays a noticeable temperature gradient, with higher temperatures observed in the top layer containing nanoparticles and lower temperatures in the bottom layer without nanoparticles. This non-uniform temperature distribution is likely due to the limited thermal interaction between the nanoparticles and the PCM

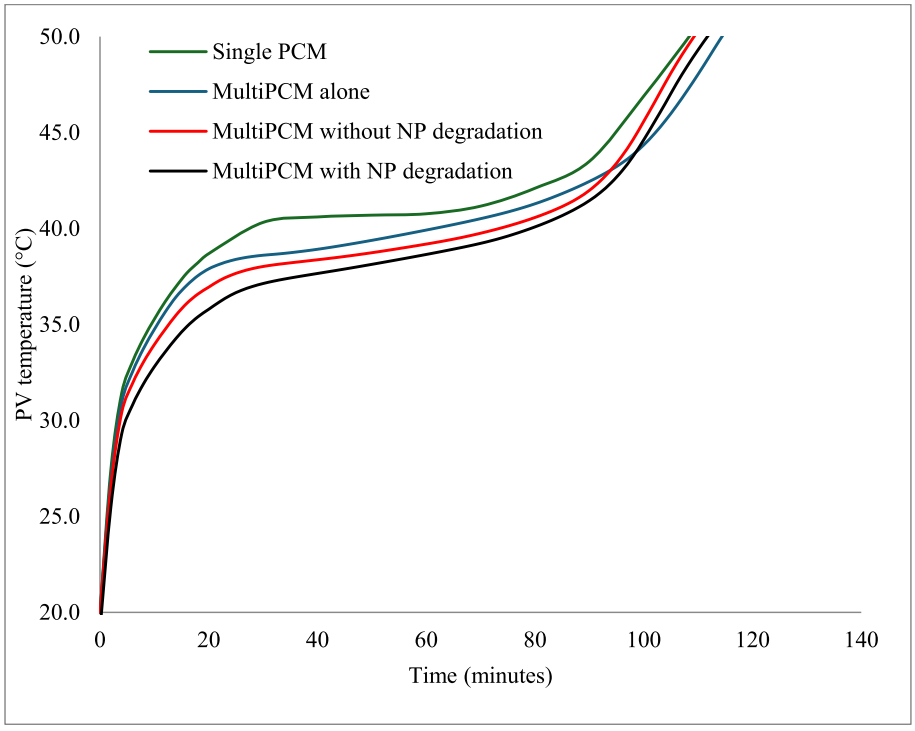


Fig. 10. Transient PV temperature profiles for two-layer PCM composite with and without degraded nanoparticle concentrations.

regions farther away from the heat source. In contrast, the two-layer PCM with degraded NP concentrations exhibits a more uniform temperature profile throughout the entire PCM domain at the 60-minute mark.

As time progresses to the 120-minute mark, the distinctions in thermal behavior among the different configurations become more pronounced. The single-layer PCM case displays a larger high-temperature region, indicating further localized heat accumulation and inefficient heat dissipation. The two-layer PCM without NPs shows improved heat distribution compared to the single-layer case, but the temperature gradient is still evident. The non-degraded NP concentration case exhibits a more pronounced temperature gradient at the 120-minute mark, with the top layer containing nanoparticles reaching significantly higher temperatures compared to the bottom layer without nanoparticles. Meanwhile, the degraded NP concentration case continues to demonstrate superior heat distribution and temperature uniformity throughout the PCM domain. Quantitively, at the 120-minute mark, the degraded NP concentration case exhibits a maximum temperature of 51 °C, which is 12 °C lower than the single-layer PCM case (63 °C), 8 °C lower than the two-layer PCM without NPs (58 °C), and 5 °C lower than the non-degraded NP case (55 °C). This indicates that the strategic grading of nanoparticle concentrations effectively mitigates the limitation observed in the uniform nanoparticle concentration case, leading to more efficient heat transfer and utilization of the PCM's latent heat storage capacity.

Fig. 10 compares the PV temperature profiles for different cases involving a two-layer multi-layer PCM arrangement over 120 min. At the start, all cases exhibit similar initial temperature of around 25 °C. As heating commences, temperature rises gradually for all cases. By 60 min, the benefits of integrating multi-layer PCMs are evident. The single PCM case witnesses the highest temperature rise to 35 °C, while the two-layer multi-layer PCM case alone shows marginally better performance at 33.5 °C. Incorporating nanoparticles further enhances heat absorption - the non-degrading arrangement records a temperature of 32 °C. Impressively, the degraded concentration arrangement displays the best thermal regulation with a lowest PV temperature of 31 °C among all cases. At a later stage of 120 min, the distinctions are more apparent. The single PCM, multi-layer PCM, non-degrading and degrading configurations register temperatures of 43 °C, 41 °C, 39.5 °C and 38 °C respectively. This clearly demonstrates that combining multi-layer PCMs alone provides thermal improvements over a single PCM. Meanwhile, integrating nanoparticles adds further gains, with the degraded degrading concentrations exhibiting the best heat diffusion characteristics consistently over time. Notably, the temperature increment gradient is also lowest for the degraded case, suggesting most uniform temperature distribution is achieved within the PV-PCM system using this approach.

Fig. 11 compares the electrical efficiency and PV temperature profiles for different cases involving a two-layer multi-layer PCM arrangement over 120 min of heat absorption. In the beginning, all cases exhibit similar initial electrical efficiency and PV temperature of around 17.5 % and 25 °C respectively. As time progresses, efficiency starts declining and temperature rising gradually for all cases. By 60 min, clear distinctions in thermal management performance are observed. The single PCM case has the lowest efficiency of 16.2 % and highest temperature of 35 °C. The three-layer multi-layer PCM arrangement alone leads to marginal improvements. Integrating nanoparticles enhances efficiency - the non-degrading case attains 16.4 % efficiency at 32.5 °C temperature. Impressively, the degraded concentration arrangement achieves the highest efficiency of 16.6 % maintaining best temperature control of 31 °C. At 100 min, the separation is most distinct. While efficiency drops to 15.5 % with a temperature of 43 °C for the single PCM case; the degraded arrangement preserves the top efficiency of 15.9 % with the lowest temperature rise of only 38 °C. This represents an improvement of around 2.7 % compared to the single PCM case, which underlines the superior thermal regulation enabled by the degraded nanoparticles-multi-layer PCMs strategy. In summary, the figure clearly demonstrates that combining three PCM layers with tailored melting points and

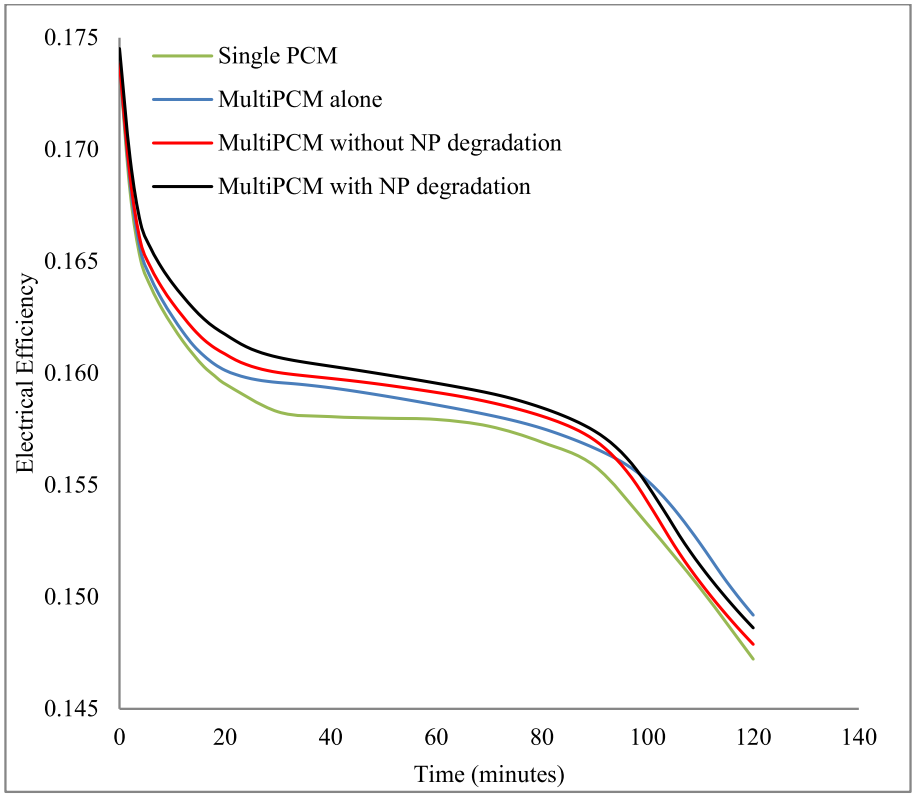


Fig. 11. Transient electrical efficiency profiles of PVT system for two-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

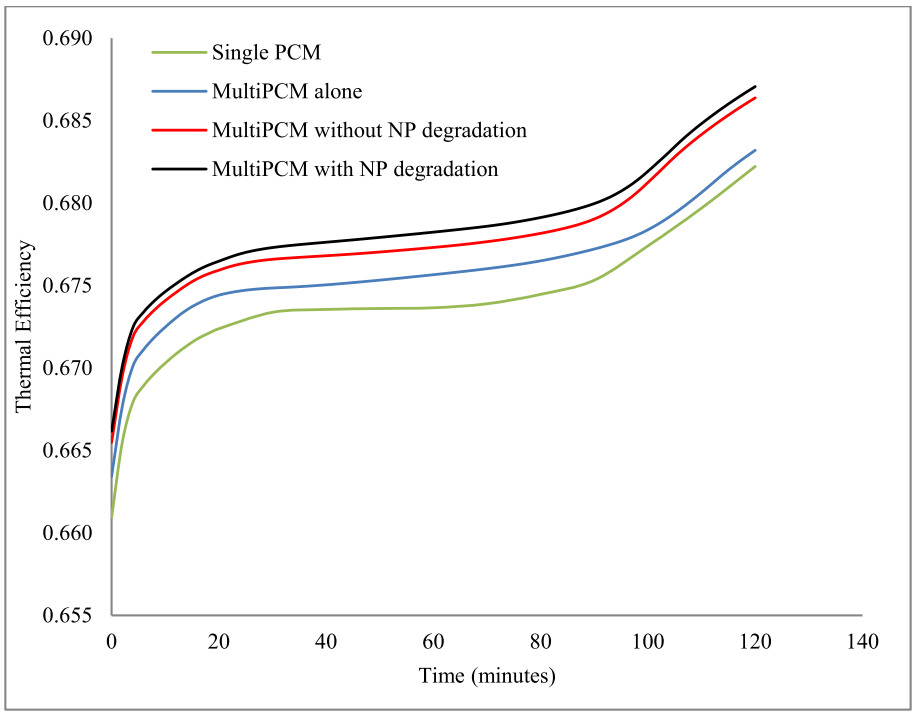


Fig. 12. Transient thermal efficiency profiles of PVT system for two-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

degraded nanoparticles concentrations in a degraded fashion offers consistent and notable improvements in maintaining both electrical efficiency and thermal behavior of the PVT system over extended periods of continuous heat accumulation compared to other alternatives investigated.

Fig. 12 compares the transient thermal efficiency profiles for different cases involving a two-layer multi-layer PCM composite with and without degraded nanoparticle concentrations. Initially, all cases exhibit similar thermal efficiency values of around 0.66. As time progresses to 60 min, the distinctions become more apparent. The single PCM case exhibits a thermal efficiency of approximately 0.674, while the multi-layer PCM case without nanoparticles shows a marginally higher efficiency of around 0.676. The non-degraded nanoparticle concentration case achieves an efficiency of about 0.677, representing an improvement of around 4.7 % compared to the single PCM case. Notably, the degraded nanoparticle concentration arrangement demonstrates the highest thermal efficiency of approximately 0.678 at this time, outperforming the single PCM case by nearly 5.8 %. Towards the end of melting around 120 min, the gap between the cases widens further. The single PCM case exhibits a thermal efficiency of around 0.677, while the multi-layer PCM without nanoparticles reaches approximately 0.678. The non-degraded nanoparticle concentration

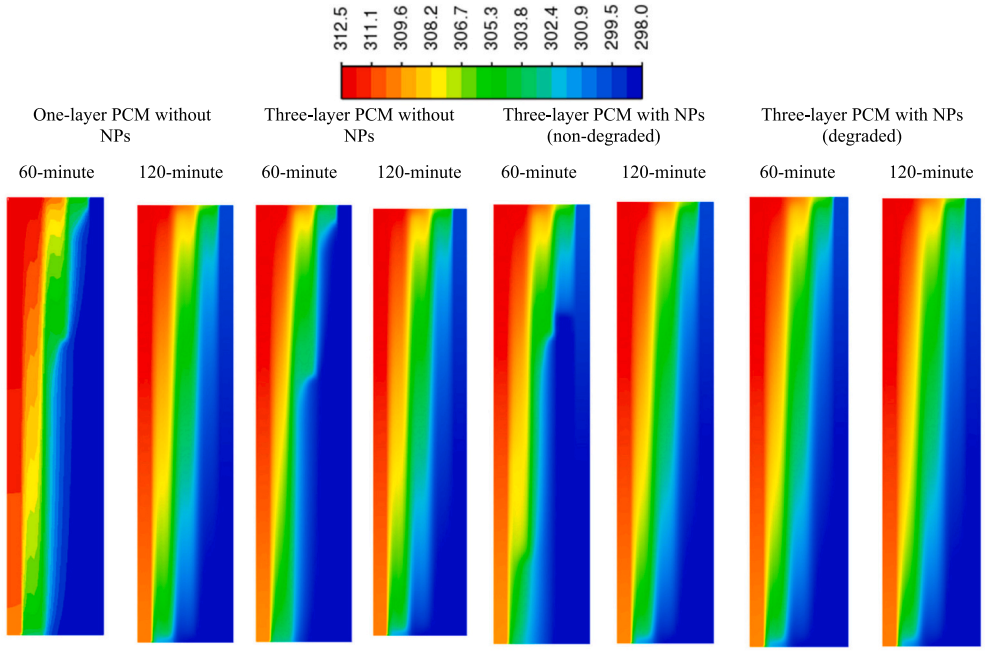


Fig. 13. Temperature distribution contours of PVT system for three-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

case achieves an efficiency of about 0.679, representing an improvement of around 5.2 % compared to the single PCM case. Impressively, the degraded nanoparticle concentration arrangement maintains the highest thermal efficiency of approximately 0.78, outperforming the single PCM case by nearly 6.7 %.

Fig. 13 shows temperature contours for four configurations of a three-layer PCM composite in a PVT system at 60- and 120-minute time instants: A single PCM layer, three PCM layers without nanoparticles, three PCM layers with uniform NP dispersion, and three PCM layers with degraded NP concentrations. Similar to the trends in two-layer configuration, the three-layer cases show more uniform distributions, with the degraded nanoparticle configuration having the most uniform temperature profile compared to the single PCM layer, which exhibits localized high temperatures near the PV sheet. At the 120-minute mark, the degraded NP concentration case quantitively exhibits a maximum temperature of 50 °C, which is 13 °C lower than the single-layer PCM case (63 °C), 7 °C lower than the three-layer PCM without NPs (57 °C), and 3 °C lower than the non-degraded NP case (53 °C). The degraded nanoparticle configuration maintains superior temperature uniformity across the PCM domain. Overall, the degraded nanoparticle arrangement enables more effective heat distribution and thermal regulation in the three-layer PCM composite for the PVT system.

Fig. 14 shows the comparative PV temperature profiles for different cases involving a three-layer multi-layer PCM arrangement from the beginning till 120 min of the melting process. In the initial 20 min, all cases exhibit similar temperatures around 25 °C. As time passes, the temperature starts increasing gradually for each case due to heat absorption. By 40 min, the benefits of multi-layer PCMs and nanoparticles become apparent. The single PCM case witnesses the highest rise to 30 °C, while the non-degrading arrangement records a temperature of 28.5 °C. Impressively, the degraded concentration configuration displays the best control, maintaining the lowest temperature of 27.5 °C. At 60 min, the deviations are more visible. The single PCM, multi-layer PCM alone and non-degraded cases register temperatures of 34 °C, 32 °C and 31 °C respectively. In contrast, the degraded arrangement continues demonstrating superior regulation, upholding the temperature at only 30 °C.

Towards the later stage of 100 min, the differences are quite discernible. The single PCM arrangement reaches the maximum temperature of 41 °C. The multi-layer PCM and non-degraded configurations record 39 °C and 38 °C respectively. Meanwhile, the degraded nanoparticle approach demonstrates optimal thermal behavior, maintaining the lowest PV temperature rise to 37 °C. This performance validates the effectiveness of our proposed hybrid enhancement approach in maintaining a uniform temperature profile for the PVT system over an extended period. However, at 120 min, a significant shift in thermal behavior is observed. Contrary to earlier time points, the degraded approach is not the best performing configuration at this final stage. This shift can be attributed to the different storage capacities and melting dynamics of the various PCM configurations. The degraded nanoparticle approach initially provides superior temperature regulation due to enhanced thermal conductivity. However, this enhanced performance also leads to faster PCM melting completion, reducing its thermal regulation capacity in the very late stages of the process. These findings highlight the complexity of thermal management in PVT systems and the importance of considering both short-term and long-term thermal management when designing such systems. They also suggest that an optimal design might involve a combination of strategies, such as using nano-enhanced PCMs for rapid initial heat dissipation coupled with a multi-PCM structure for sustained long-term thermal regulation.

Fig. 15 compares the transient electrical efficiency profiles for different cases involving a three-layer multi-layer PCM arrangement over 120 min. Initially, all cases exhibit similar efficiencies around 17.5 % when temperature is uniform. As time passes, efficiency begins decreasing steadily for each case due to rising temperatures. By 60 min, the impact of thermal management variations is evident. The single PCM case witnesses the maximum drop in efficiency to 16.2 %. Multi-layer PCMs alone provide marginal improvement, with the non-degrading approach achieving 16.3 % efficiency. Remarkably, the degraded concentration configuration retains the highest efficiency of 16.5 % among

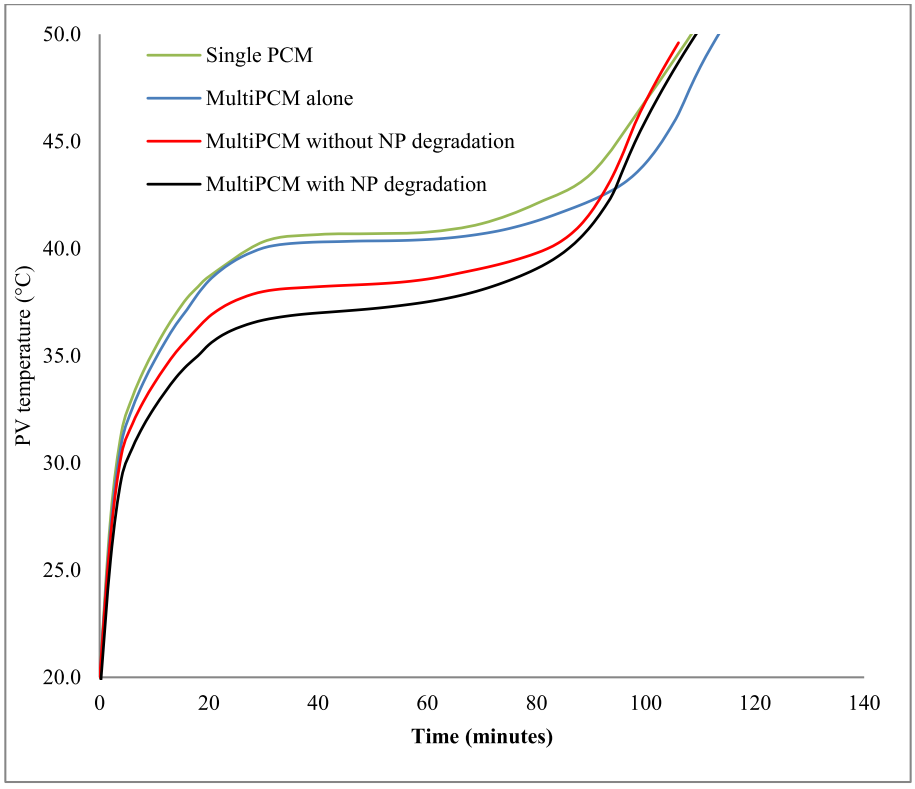


Fig. 14. Transient temperature profiles of PV module for three-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

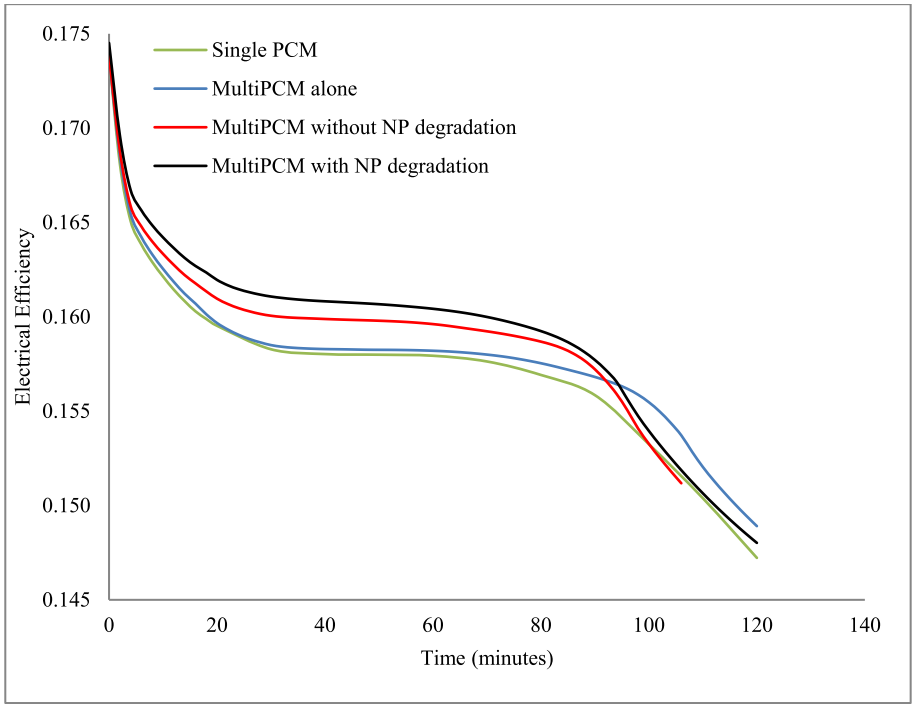


Fig. 15. Transient electrical efficiency profiles of PVT system for three-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

all cases at this point.

Up to around 100 min, the deviations are more discernible. The efficiency drops to 15.5 % for the single PCM arrangement. The multi-layer PCM and non-degraded cases record 15.7 % and 15.9 % efficiencies respectively. The degraded nanoparticles strategy in the multi-layer PCM system demonstrates superior performance, maintaining the highest electrical efficiency of 16.1 %. This performance underscores the effectiveness of the proposed approach in sustaining electrical efficiency over extended periods of heat accumulation. However, at 120 min, an interesting shift in performance can be observed. The multi-PCM with NP degradation case is no longer the best performing configuration at this final time point. This unexpected result provides valuable insights into the long-term behavior of these complex thermal management systems. This shift in performance can be attributed to the interplay between enhanced heat transfer and PCM melting dynamics. The nano-enhanced PCMs initially provide superior heat dissipation, maintaining higher electrical efficiencies. However, their accelerated melting may lead to earlier saturation of their thermal storage capacity, reducing their effectiveness in later stages. These findings suggest that an optimal thermal management strategy for PVT systems may need to be more sophisticated than a single, static configuration. While the current study provides crucial insights, it also points to the need for further investigation to explore hybrid designs that combine the strengths of different approaches to maintain optimal performance throughout the entire operational period.

In summary, the hybrid enhancement approach validates its effectiveness in maximizing both thermal regulation and electrical production of the integrated PVT system over around 100 min. The degraded nanoparticles-multi-layer PCM configuration achieves the best regulation of PV module temperature and preservation of electrical efficiency gains consistently throughout the entire PCM melting process.

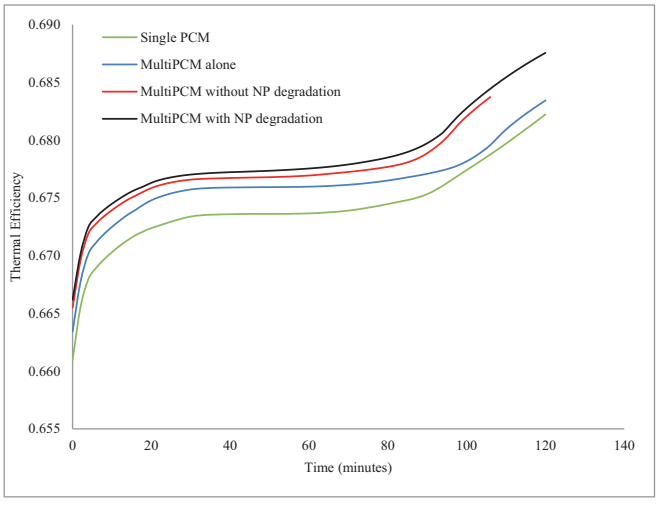


Fig. 16. Transient thermal efficiency profiles of PVT system for three-layer multi-layer PCM composite with and without degraded nanoparticle concentrations.

Fig. 16 compares the transient thermal efficiency profiles of the PVT system for a three-layer multi-layer nano-PCM composite with and without degraded nanoparticle concentrations. Initially, all cases exhibit similar thermal efficiency values of around 0.665, indicating comparable performance at the outset of the melting mode. As time progresses to 60 min, distinctions in thermal efficiency become more apparent among the different configurations. The case involving a single PCM exhibits a thermal efficiency of about 0.674 at 60 min. The multi-layer PCM case without nanoparticles shows a marginally higher efficiency of around 0.676, representing a slight improvement compared to the single PCM case. This can be attributed to the presence of multiple PCMs with different melting points, which enhances the heat absorption capabilities of the system. The non-degraded NP concentration case, where nanoparticles are uniformly dispersed throughout the multi-layer PCM composite, achieves a thermal efficiency of about 0.676 at 60 min. This configuration outperforms the single PCM case by approximately 3.7 %, highlighting the benefits of incorporating nanoparticles to enhance the thermal conductivity of the PCM composite. Meanwhile, the degraded NP concentration arrangement demonstrates the highest thermal efficiency of approximately 0.678 at 60 min. This configuration outperforms the single PCM case by nearly 5.4 %, indicating the superior

thermal performance achieved by strategically grading the nanoparticle concentrations within the multi-layer PCM composite. As the melting process progresses towards the end, around 120 min, the gap between the different cases widens further. The single PCM case exhibits a thermal efficiency of around 0.677, while the multi-layer PCM without nanoparticles reaches approximately 0.678. The non-degraded nanoparticle concentration case achieves an efficiency of about 0.680, representing an improvement of around 5.7 % compared to the single PCM case. However, the degraded NP concentration arrangement maintains the highest thermal efficiency of approximately 0.682 at 120 min. This configuration outperforms the single PCM case by nearly 6.7 %, demonstrating its superiority in terms of thermal performance and energy storage capabilities over an extended duration.

The observed trend in electrical and thermal efficiency profiles highlights the advantages of employing a degraded nanoparticle concentration within a multi-layer PCM composite for the PVT system. By strategically arranging layers of PCMs with varying melting points and tailoring the nanoparticle concentrations in a degraded fashion, more uniform heat distribution and enhanced thermal interaction between the nanoparticles and PCM regions can be achieved. Overall, this approach consistently exhibits the highest thermal efficiency throughout the melting process, outperforming the other configurations investigated. This innovative thermal management strategy enables more efficient heat absorption, temperature regulation, and energy storage capabilities within the PVT system, highlighting its potential for optimizing thermal energy management in various energy applications.

In summary, the three-layer configuration with degraded nanoparticle concentrations achieves superior performance due to its improved heat distribution. The highest nanoparticle concentration near the heat source (PV module) enhances initial heat absorption and distribution. The middle layer with moderate concentration helps maintain efficient heat transfer, while the bottom layer with no nanoparticles acts as a thermal buffer preventing rapid heat loss to the environment. This arrangement allows for more uniform melting and better overall thermal regulation compared to single-layer or two-layer configurations, as evidenced by the temperature contours in Fig. 13 and the performance metrics in Figs. 14–16.

Compared to conventional PVT thermal management methods like air or water circulation, the proposed multi-layer nano-PCM approach offers several advantages. It requires no pumping power, reducing parasitic energy losses. The system is also simpler, with fewer moving parts, potentially increasing reliability and reducing maintenance needs. While some studies have explored uniform nano-PCMs or simple multi-layer PCMs, the degraded concentration approach uniquely addresses the limitation of decreasing thermal interaction in regions far from the heat source, potentially offering superior performance to these existing strategies.

5.4. Limitations and future considerations

While PVT systems are typically operated for 6–8 h daily [46], initial simulations in this study were focused on a two-hour period. This timeframe was chosen to capture critical transient behavior during the system's initial response to thermal loads. Key differences in heat absorption rates, temperature regulation, and the onset of phase change processes between configurations were revealed through this approach. With the current 21-mm thick PCM layer, complete melting was predicted by the model to occur around the two-hour mark, allowing the full melting cycle and its effects on thermal management to be analyzed.

The current study is constrained by computational resources required for modeling complex phase change dynamics and nanoparticle-enhanced heat transfer over extended periods. Additionally, thicker PCM layers are necessitated for longer operational durations to maintain effective thermal management, further increasing computational demands. The need for further research to fully understand the long-term performance and practical applicability of the proposed multi-layer nano-modified PCM configuration is highlighted by these limitations. Future research directions also could include developing cost-effective manufacturing processes for multi-layer nano-PCMs, conducting long-term stability and durability studies, and investigating the scalability of the system for various applications.

6. Conclusions

This study investigated a novel thermal management approach for PVT systems using multi-layer nano-PCMs with degraded nanoparticle concentrations. Through comprehensive numerical simulations, the energy conversion and thermal performance were investigated for three distinct configurations: a single-layer uniform nano-PCM, a two-layer nano-PCM with graded concentrations, and a three-layer nano-PCM with degraded nanoparticle concentrations. The main conclusions are:

1. Among various nanoparticle materials, graphene-based nano-PCM configuration exhibited superior thermal regulation capabilities, maintaining the lowest PV module temperatures and increasing electrical efficiencies by up to 4.1 % compared to pure PCM after 1 h of operation.
2. The two-layer degraded nano-PCM configuration demonstrated improved performance over the uniform case, keeping PV temperatures lower and electrical efficiencies up to 2.7 % higher throughout the 100-minute melting duration.
3. The three-layer degraded nano-PCM arrangement consistently demonstrated the best performance, maintaining the lowest PV temperatures (up to 5 °C lower) and highest electrical efficiencies (up to 4 % higher) throughout the PCM melting duration.
4. The three-layer degraded configuration exhibited the highest thermal efficiency of approximately 0.682 at 100 min, outperforming the single PCM case by nearly 6.7 %. Furthermore, this configuration consistently upheld the optimal electrical efficiency of 16.1 % over the extended 100-minute duration, compared to 15.5 % for the single PCM case.
5. The combination of multi-layer PCMs with strategically degraded nanoparticle loadings enabled more uniform heat distribution within the nano-PCM composite. This approach can successfully address the limitation of decreasing thermal interaction between nanoparticles and PCM regions farther from the heat source (PV cell), which is observed in uniform nano-PCMs.

These findings provide valuable insights for optimizing thermal management in PVT systems. The improved thermal and electrical efficiency of the proposed PVT system would have significant implications for real-world applications. The improved electrical efficiency could translate to substantial energy savings over the system's lifetime. Moreover, better temperature regulation extends the lifespan of PV cells by reducing thermal stress. The enhanced thermal efficiency also increases the potential for utilizing waste heat in applications such as water heating or space heating, improving overall energy utilization in residential or commercial settings.

CRediT authorship contribution statement

Saleh Al Arni: Writing – original draft, Conceptualization. **Jasim M. Mahdi:** Writing – review & editing, Conceptualization. **Azher M. Abed:** Writing – original draft, Investigation. **Karrar A. Hammoodi:** Investigation, Formal analysis. **Husam Abdulrasool Hasan:** Writing – original draft, Methodology. **Raad Z. Homod:** Writing – review & editing, Supervision. **Nidhal Ben Khedher:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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